

NATNAEL GETACHEW

FULL-STACK SOFTWARE ENGINEER | SYSTEMS & DISTRIBUTED WEB

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PROFESSIONAL SUMMARY

Computer Engineering graduate & IBT College Software Development trainee specializing in Next.js App Router, React 18, TypeScript, Python architecture, and native Android solutions. Passionate about scalable distributed systems, sub-100ms algorithmic latency, and clean test-driven architectures.

CORE TECHNICAL SKILLS

Frontend: Next.js 14 (App Router), React 18, TypeScript, Tailwind CSS, Zustand, Recharts, HTML5, CSS3

Backend & DB: NestJS, Node.js, Express, PostgreSQL, Prisma ORM, REST APIs, TeleBirr Gateway, JWT Auth

Architecture: Python 3 OOP, Directed Graph Routing, Factory & Singleton Patterns, unittest, SQLite

Mobile & Tools: Android Studio, Java/Kotlin, RecyclerView, Git & GitHub PR reviews, Linux/Bash, CI/CD

PROFESSIONAL EXPERIENCE & LEADERSHIP

Full-Stack Trainee & Project Lead - IBT Academy Cohort (2024 - Present)

- Directing team deliverables for full-stack web products, conducting PR code reviews and architectural audits.
- Implemented test suites and enforced strict TypeScript typing and React 18 concurrent UI patterns.

Android Developer Intern - M.A.D Technologies, Addis Ababa (2022 - 2023)

- Translated high-fidelity Figma UI wireframes into responsive Android layouts with ViewBinding & RecyclerView.
- Implemented offline-first SQLite database synchronization, significantly reducing app startup latency.
- Solved memory leaks and debugged edge-case race conditions in background async tasks.

FEATURED PROJECTS

Addis Eats - Next-Gen Food Ordering & TeleBirr Gateway [Next.js, TypeScript, Zustand, TeleBirr]

- Engineered high-velocity ordering platform with sub-city delivery fee calculation and Ethiopian phone validation.
- Architected Zustand cart state engine with inventory-hard ceilings and atomic checkout synchronization.

AddisBank - Core Banking & Liquidity Transfer Graph [Python 3.11, Graphs, Design Patterns]

- Designed multi-tier account engine with directed graph inter-branch liquidity routing and zero balance leakage.
- Achieved 100% test assertion coverage using unittest for concurrent transfer limits and invariant checks.

Native Android Lexicon & Offline Dictionary [Android Studio, Java/Kotlin, SQLite]

- Built sub-100ms offline terminology search indexing using debounced SQLite queries and custom RecyclerView diffing.

EDUCATION & CREDENTIALS

Advanced Digital Skills: Software Development & QA - IBT College of Canada (2024 - Present)

- Specialized in modern web architectures (Next.js, React, TypeScript), QA test automation & CI/CD pipelines.

BSc in Computer Engineering - Debre Birhan University, Ethiopia (2017 - 2022)

- 5-year engineering foundation in DSA Algorithms, Distributed Systems, Microprocessors, and Database Systems.